Combinatorial Analysis

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Chapter 1

Ordinary Generating Functions

1.1 Formal Power Series and Ordinary Generating Functions

Consider tilings of a $2 \times n$ rectangle using 2×1 dominoes. Let a_n be the number of such tilings. Consider the domino covering the upper-left corner. It is either vertical, leaving a $2 \times (n-1)$ rectangle, or horizontal; in the latter case the lower-left corner must also be covered horizontally, leaving a $2 \times (n-2)$ rectangle.

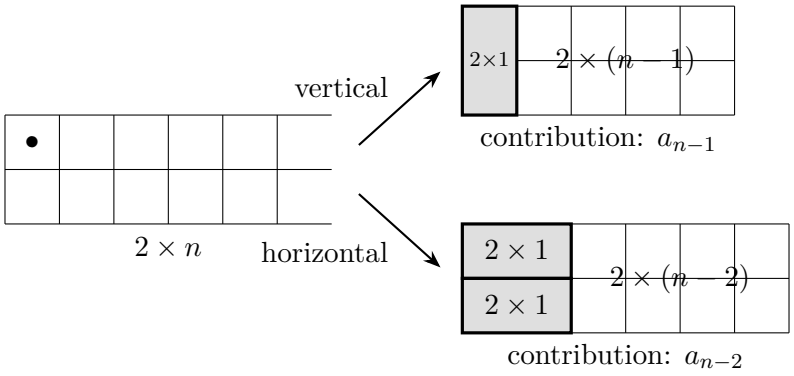


Figure 1: Decomposition of a $2 \times n$ domino tiling according to the domino covering the upper-left corner.

The two cases are disjoint and exhaustive. Therefore

$$\boxed{a_n = a_{n-1} + a_{n-2}, \qquad a_0 = a_1 = 1.}$$

Thus the tiling numbers are the shifted Fibonacci numbers,

$$a_n = F_{n+1}.$$

This recurrence determines the sequence efficiently, but it does not give a_n directly. We can also count the same tilings without recursion. If a tiling contains k pairs of horizontal dominoes, then it contains $n - 2k$ vertical dominoes. Treating each horizontal pair as a block of width 2 and each vertical domino as a block of width 1, we arrange k blocks of the first type and $n - 2k$ of the second type. Hence

$$a_n = \sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n-k}{k}.$$

This is an explicit finite sum, although it is still not especially convenient for computation or asymptotic analysis.
Solving the recurrence gives the closed form

$$a_n = \frac{1}{\sqrt{5}} \left(\phi^{n+1} - \psi^{n+1} \right), \qquad \phi = \frac{1+\sqrt{5}}{2}, \qquad \psi = \frac{1-\sqrt{5}}{2}.$$

Since $|\psi| < 1$,

$$a_n \sim \frac{1}{\sqrt{5}} \phi^{n+1}.$$

These are different descriptions of the same counting sequence: the recurrence reflects its recursive combinatorial structure, the finite sum gives a direct enumeration, and the closed form makes its growth transparent. Generating functions provide another representation: instead of describing the numbers $a_0, a_1, a_2, \ldots$ separately, we encode the entire sequence as the coefficients of a single formal power series.

Definition 1.1.1 (Ordinary Generating Function)**.** For a sequence $(a_n)_{n\geq 0}$, its *ordinary generating function (OGF)* is the formal power series

$$A(x) := \sum_{n\geq 0} a_n x^n.$$

For any formal power series $A(x) = \sum_{k\geq 0} a_k x^k$, the notation $[x^n]A(x) := a_n$ denotes the coefficient of x^n in $A(x)$.

Example 1.1.2 (Domino Tilings)**.** For the domino-tiling sequence,

$$A(x) = a_0 + a_1 x + \sum_{n\geq 2}(a_{n-1} + a_{n-2})x^n = 1 + x + x(A(x) - 1) + x^2 A(x),$$

and therefore

$$\boxed{A(x) = \frac{1}{1 - x - x^2}.}$$

Example 1.1.3 (Counting Fruits)**.** Suppose one may buy at most one apple and at most two bananas. Then

$$(1 + x)(1 + x + x^2)$$

has $[x^n]$ equal to the number of choices containing n fruits.

Example 1.1.4 (Encoding Total Price)**.** If the apple and banana have prices 2 and 3, respectively, then

$$(1 + y^2)(1 + y^3 + y^6)$$

has $[y^n]$ equal to the number of choices having total price n.

Definition 1.1.5 (Formal Power Series)**.** Let R be a commutative ring with identity. The ring of formal power series over R is

$$R[[x]] := \left\{ \sum_{n\geq 0} a_n x^n : a_n \in R \right\}.$$

Addition is coefficientwise, while multiplication is defined by convolution:

$$\left(\sum_{n\geq 0} a_n x^n\right)\left(\sum_{n\geq 0} b_n x^n\right) = \sum_{n\geq 0} c_n x^n, \qquad c_n = \sum_{i=0}^{n} a_i b_{n-i}.$$

The variable x is treated as a formal indeterminate; no numerical value of x or analytic convergence is required.

Proposition 1.1.6. A formal power series $A(x) = \sum_{n\geq 0} a_n x^n \in R[[x]]$ is a unit in $R[[x]]$ if and only if a_0 is a unit in R.

Proof. Suppose $B(x) = \sum_{n\geq 0} b_n x^n$ satisfies $A(x)B(x) = 1$. Comparing coefficients gives

$$a_0 b_0 = 1, \qquad a_1 b_0 + a_0 b_1 = 0, \qquad a_2 b_0 + a_1 b_1 + a_0 b_2 = 0, \ldots$$

Thus a_0 must be a unit. Conversely, if a_0 is a unit, then $b_0 = a_0^{-1}$, and each subsequent b_n is determined recursively by the coefficient equation for x^n. □

For example, $1 - x - x^2$ has constant coefficient 1, so it is a unit in $\mathbb{C}[[x]]$ and

$$\frac{1}{1 - x - x^2} \in \mathbb{C}[[x]]$$

is well-defined.

Definition 1.1.7 (Convergence in $R[[x]]$)**.** A sequence $A_1(x), A_2(x), \ldots$ converges to $A(x) = \sum_{n \geq 0} a_n x^n$ in $R[[x]]$ if, for every fixed $n \geq 0$, there exists N such that

$$[x^n]A_j(x) = a_n \qquad \text{for all } j > N.$$

Thus every fixed coefficient must eventually stabilize; merely converging numerically is not sufficient. For example, analytically,

$$e^{x+1} = \lim_{N \to \infty} \sum_{m=0}^{N} \frac{(x+1)^m}{m!}.$$

However, for fixed k,

$$[x^k] \sum_{m=0}^{N} \frac{(x+1)^m}{m!} = \frac{1}{k!} \sum_{m=0}^{N-k} \frac{1}{m!},$$

which approaches $e/k!$ but never becomes eventually constant. Hence this sequence does not converge in $\mathbb{C}[[x]]$ in the formal sense. Instead, as a formal power series,

$$e^{x+1} = e\, e^x = e \sum_{n \geq 0} \frac{x^n}{n!}.$$

For $A(x) = \sum_{n \geq 0} a_n x^n \in R[[x]]$, its *formal derivative* is defined coefficientwise by

$$\boxed{A'(x) := \frac{d}{dx} A(x) = \sum_{n \geq 1} n a_n x^{n-1}.}$$

This is an algebraic definition and does not require analytic convergence.

1.2 Combinatorial Constructions

A *combinatorial class* $\mathcal{A}$ is a set of combinatorial objects equipped with a notion of size. Let $\mathcal{A}_n$ denote the set of objects of size n, and let

$$a_n := |\mathcal{A}_n|, \qquad A(x) := \sum_{n \geq 0} a_n x^n.$$

Proposition 1.2.1 (Disjoint Union)**.** If $\mathcal{C} = \mathcal{A} \sqcup \mathcal{B}$, where $\sqcup$ denotes disjoint union, then

$$\boxed{C(x) = A(x) + B(x).}$$

Indeed, a $\mathcal{C}$-object of size n is either an $\mathcal{A}$-object or a $\mathcal{B}$-object of size n, so $c_n = a_n + b_n$.

Proposition 1.2.2 (Product)**.** If $\mathcal{C} = \mathcal{A} \times \mathcal{B}$ and the size of a pair is the sum of the sizes of its two components, then

$$c_n = \sum_{i=0}^{n} a_i b_{n-i}, \qquad \boxed{C(x) = A(x)B(x).}$$

Thus multiplication of OGFs corresponds exactly to convolution of their coefficient sequences.

Definition 1.2.3 (Sequence Construction)**.** For a combinatorial class $\mathcal{B}$, $\mathrm{Seq}(\mathcal{B})$ denotes the class of all finite ordered sequences of objects from $\mathcal{B}$.

Assume $|\mathcal{B}_0| = 0$, so that every $\mathcal{B}$-object has positive size. A sequence containing exactly k objects has OGF $B(x)^k$. Hence

$$\boxed{\mathrm{Seq}(\mathcal{B}) \quad \longleftrightarrow \quad \sum_{k\geq 0} B(x)^k = \frac{1}{1-B(x)}.}$$

Definition 1.2.4 (Composition of Formal Power Series)**.** Let $A(x) = \sum_{n\geq 0} a_n x^n$ and $B(x) = \sum_{n\geq 0} b_n x^n \in R[[x]]$, and assume $b_0 = 0$. The *composition* of A and B is

$$\boxed{A(B(x)) := \sum_{n\geq 0} a_n (B(x))^n.}$$

The condition $b_0 = 0$ ensures that the coefficient of each fixed power x^m receives contributions from only finitely many terms, so the composition is well-defined in $R[[x]]$. In particular, taking $A(x) = 1/(1-x)$ gives

$$A(B(x)) = \frac{1}{1-B(x)} = \sum_{k\geq 0} B(x)^k,$$

which is exactly the generating function for $\mathrm{Seq}(\mathcal{B})$.

1.3 Partitions and Compositions

Definition 1.3.1 (Integer Partition)**.** An *integer partition* of n is a way to write n as a sum of positive integers, where the order of the parts does not matter. Let p_n denote the number of integer partitions of n.

Example 1.3.2. For $n = 15$,

$$15 = 5 + 4 + 3 + 2 + 1$$

is an integer partition of 15. Since the order of the parts does not matter,

$$5 + 4 + 3 + 2 + 1 \qquad \text{and} \qquad 1 + 4 + 2 + 5 + 3$$

represent the same integer partition.

The OGF for integer partitions is

$$P(x) := \sum_{n\geq 0} p_n x^n.$$

For each possible part size k, that part may occur $0, 1, 2, \ldots$ times, contributing

$$1 + x^k + x^{2k} + \cdots = \frac{1}{1-x^k}.$$

Since the multiplicities of different part sizes may be chosen independently,

$$\boxed{P(x) = \prod_{k\geq 1} \frac{1}{1-x^k}.}$$

If all parts are required to be at most m, then

$$P_m(x) = \prod_{k=1}^{m} \frac{1}{1-x^k}.$$

Definition 1.3.3 (Composition)**.** A *composition* of n is an ordered sequence of positive integers whose sum is n. Thus, unlike an integer partition, the order of the parts matters.

Example 1.3.4. For $n = 15$,

$$(5, 4, 3, 2, 1) \qquad \text{and} \qquad (1, 4, 2, 5, 3)$$

are two different compositions of 15, since the order of the parts matters.

Let $\mathcal{B} = \{1, 2, 3, \ldots\}$ contain one object of each positive size. Then

$$B(x) = x + x^2 + x^3 + \cdots = \frac{x}{1-x}.$$

Since a composition is precisely a finite sequence of positive parts,

$$C(x) = \frac{1}{1-B(x)} = \frac{1-x}{1-2x} = 1 + \sum_{n\geq 1} 2^{n-1}x^n.$$

Hence

$$\boxed{\#\{\text{compositions of } n\} = 2^{n-1}, \qquad n \geq 1.}$$

Definition 1.3.5 (Set Partition)**.** For $n \geq 1$, write $[n] := \{1, 2, \ldots, n\}$. A *set partition* of $[n]$ is a collection of nonempty pairwise disjoint subsets, called *blocks*, whose union is $[n]$.

Definition 1.3.6 (Stirling Number of the Second Kind)**.** The *Stirling number of the second kind* $S(n, k)$ is the number of partitions of $[n]$ into exactly k nonempty blocks.

Considering the block containing n, either n forms a new singleton block or it joins one of the k existing blocks. Therefore

$$\boxed{S(n, k) = S(n-1, k-1) + kS(n-1, k),}$$

with $S(0, 0) = 1$ and $S(n, 0) = 0$ for $n \geq 1$.
For fixed k, define the OGF

$$G_k(x) := \sum_{n\geq k} S(n, k)x^n.$$

The recurrence gives $G_k(x) = xG_{k-1}(x) + kxG_k(x)$, and hence

$$\boxed{G_k(x) = \frac{x}{1-kx}G_{k-1}(x) = \frac{x^k}{(1-x)(1-2x)\cdots(1-kx)}.}$$

1.4 Catalan Numbers

Definition 1.4.1 (Catalan Numbers)**.** The *Catalan numbers* are

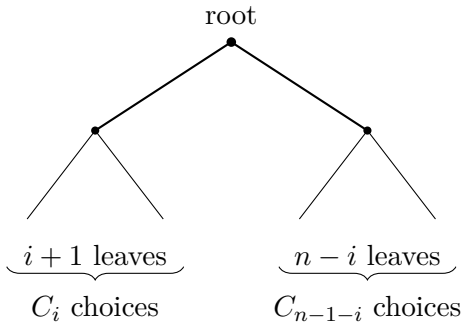
$$\boxed{C_n = \frac{1}{n+1}\binom{2n}{n}, \qquad n \geq 0.}$$

The first values are $C_0 = 1$, $C_1 = 1$, $C_2 = 2$, and $C_3 = 5$.

The same number C_n counts several combinatorial structures:

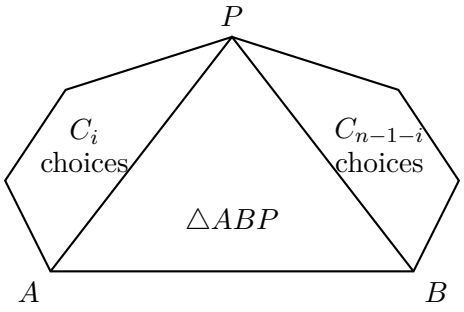
- planar binary trees with $n + 1$ leaves;
- triangulations of a convex $(n + 2)$-gon;
- lattice paths from $(0, 0)$ to (n, n) using North steps $N = (0, 1)$ and East steps $E = (1, 0)$ and staying weakly above the diagonal $y = x$.

Example 1.4.2 (Planar Binary Trees)**.** A planar binary tree with $n + 1$ leaves splits at its root into a left subtree with $i + 1$ leaves and a right subtree with $n - i$ leaves.



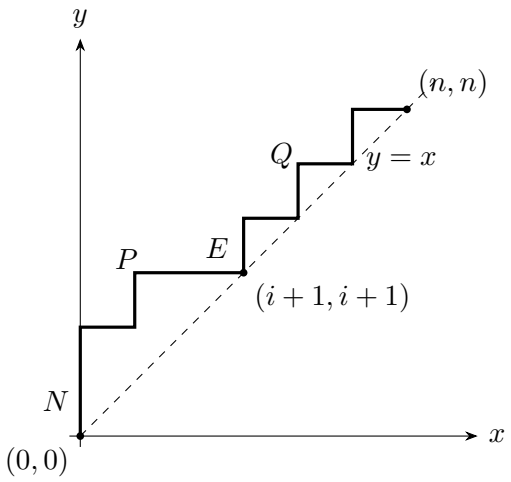
Since $(i + 1) + (n - i) = n + 1$, fixing i gives $C_i C_{n-1-i}$ choices for the two subtrees. Summing over $i = 0, \ldots, n - 1$ gives the Catalan recurrence.

Example 1.4.3 (Polygon Triangulations)**.** Fix a boundary edge AB of a convex $(n+2)$-gon. In every triangulation, AB belongs to a unique triangle ABP.



If P leaves i vertices on one side of AB and $n - 1 - i$ on the other, the two remaining regions are an $(i+2)$-gon and an $(n-i+1)$-gon. They can be triangulated independently in $C_i C_{n-1-i}$ ways.

Example 1.4.4 (Lattice Paths)**.** A Catalan path of size n goes from $(0, 0)$ to (n, n) using North and East steps while staying weakly above the diagonal $y = x$.



Let $(i+1, i+1)$ be the first return to the diagonal after $(0,0)$. The path has a unique decomposition $NPEQ$, where, after translation, P is a Catalan path of size i and Q is a Catalan path of size $n-1-i$. Hence this case contributes $C_i C_{n-1-i}$.

Proposition 1.4.5 (Catalan Recurrence)**.** For $n \geq 1$,

$$\boxed{C_n = \sum_{i=0}^{n-1} C_i C_{n-1-i}.}$$

Proof. For planar binary trees, removing the root separates a tree with $n+1$ leaves into a left and a right subtree. If they contain $i+1$ and $j+1$ leaves, respectively, then $i+j = n-1$. They may be chosen independently in $C_i C_j$ ways, and summing over all $i+j = n-1$ gives the recurrence.
For polygon triangulations, fix one boundary edge of a convex $(n+2)$-gon. In every triangulation this edge belongs to a unique triangle. If the third vertex leaves i vertices on one side and $n-1-i$ on the other, the remaining two regions are an $(i+2)$-gon and an $(n-i+1)$-gon. Their triangulations may be chosen independently in $C_i C_{n-1-i}$ ways, and summing over all possible positions of the third vertex gives the same recurrence.
For lattice paths, every nonempty Catalan path starts with N. Let $(i+1, i+1)$ be its first return to the diagonal after $(0,0)$. The path has a unique decomposition $NPEQ$, where, after translation, P is a Catalan path of size i and Q is a Catalan path of size $n-1-i$. Hence there are $C_i C_{n-1-i}$ paths with this first-return position, and summing over $i = 0, \ldots, n-1$ again gives the recurrence. □

Define the OGF

$$C(x) := \sum_{n \geq 0} C_n x^n.$$

Since $C_0 = 1$, the recurrence gives

$$C(x) = 1 + \sum_{n \geq 1} \left(\sum_{i=0}^{n-1} C_i C_{n-1-i} \right) x^n.$$

By the product rule for generating functions,

$$\sum_{n \geq 1} \left(\sum_{i=0}^{n-1} C_i C_{n-1-i} \right) x^n = x C(x)^2.$$

Hence

$$\boxed{C(x) = 1 + xC(x)^2.}$$

Equivalently,

$$xC(x)^2 - C(x) + 1 = 0.$$

Solving this quadratic equation gives

$$C(x) = \frac{1 \pm \sqrt{1-4x}}{2x}.$$

The plus sign cannot be chosen, since

$$\frac{1 + \sqrt{1-4x}}{2x}$$

does not define a formal power series with constant coefficient 1. Therefore

$$\boxed{C(x) = \frac{1 - \sqrt{1-4x}}{2x}.}$$

To extract its coefficients, recall the formal binomial expansion

$$(1+z)^\alpha = \sum_{n \geq 0} \binom{\alpha}{n} z^n, \qquad \binom{\alpha}{n} := \frac{\alpha(\alpha-1)\cdots(\alpha-n+1)}{n!}.$$

Taking $\alpha = \frac{1}{2}$ and $z = -4x$,

$$\sqrt{1-4x} = \sum_{n \geq 0} \binom{1/2}{n} (-4x)^n.$$

Thus

$$C(x) = -\frac{1}{2x} \sum_{n \geq 1} \binom{1/2}{n} (-4)^n x^n.$$

Therefore

$$C_n = -\frac{1}{2} \binom{1/2}{n+1} (-4)^{n+1}.$$

Simplifying the generalized binomial coefficient gives

$$-\frac{1}{2} \binom{1/2}{n+1} (-4)^{n+1} = \frac{1}{n+1} \binom{2n}{n}.$$

Hence

$$\boxed{C_n = \frac{1}{n+1} \binom{2n}{n}.}$$

Chapter 2

Exponential Generating Functions

2.1 Exponential Generating Functions

Definition 2.1.1 (Exponential Generating Function)**.** Given a sequence $(a_n)_{n\geq 0}$, its *exponential generating function (EGF)* is

$$\boxed{A^{\exp}(x) := \sum_{n\geq 0} a_n \frac{x^n}{n!}.}$$

The difference between the ordinary and exponential generating functions of the same sequence is

$$A(x) = \sum_{n\geq 0} a_n x^n, \qquad A^{\exp}(x) = \sum_{n\geq 0} a_n \frac{x^n}{n!}.$$

A *labeled structure of size* n is a combinatorial structure built on the labeled set

$$[n] := \{1, 2, \ldots, n\},$$

so that the labels distinguish the underlying elements. The factor $1/n!$ makes EGFs particularly useful for counting labeled structures.

Example 2.1.2 (Permutations)**.** Let $a_n = n!$, the number of permutations of $[n]$. Its OGF is $\sum_{n\geq 0} n! x^n$, while its EGF has the simple form

$$A^{\exp}(x) = \sum_{n\geq 0} \frac{n!}{n!} x^n = \sum_{n\geq 0} x^n = \boxed{\frac{1}{1-x}}.$$

Example 2.1.3. If $a_n = 2^n$, then

$$A(x) = \sum_{n\geq 0} 2^n x^n = \frac{1}{1-2x}, \qquad A^{\exp}(x) = \sum_{n\geq 0} \frac{2^n}{n!} x^n = \boxed{e^{2x}}.$$

Example 2.1.4 (Circular Arrangements)**.** For $n \geq 1$, the number of arrangements of n labeled elements around a circle is $(n-1)!$. Hence, using the convention $a_0 = 1$,

$$A^{\exp}(x) = 1 + \sum_{n\geq 1} \frac{(n-1)!}{n!} x^n = 1 + \sum_{n\geq 1} \frac{x^n}{n}.$$

Since $-\log(1-x) = \sum_{n\geq 1} x^n/n$,

$$\boxed{A^{\exp}(x) = 1 - \log(1-x).}$$

2.2 Permutations and Cycles

Let S_n denote the *symmetric group* on $[n] = \{1, 2, \ldots, n\}$, i.e., the set of all permutations of $[n]$. Thus $|S_n| = n!$.

A permutation $\sigma \in S_n$ may be represented in several equivalent ways. In *two-line notation*,

$$\sigma = \begin{pmatrix} 1 & 2 & \cdots & n \\ \sigma(1) & \sigma(2) & \cdots & \sigma(n) \end{pmatrix}.$$

It may also be represented by its *function digraph*, with a directed edge $i \to \sigma(i)$ for each $i \in [n]$, or by *cycle notation*. Cyclic rotations represent the same cycle; for example,

$$(143) = (314) = (431).$$

Every permutation decomposes uniquely into disjoint cycles, up to the order of the cycles and cyclic rotation within each cycle. For a standard representation, we may write the largest element of each cycle first and order the cycles according to these largest elements.

Definition 2.2.1 (Cycle Type)**.** For $\sigma \in S_n$, let $c_i(\sigma)$ denote the number of cycles of length i in its cycle decomposition. The *cycle type* of σ is

$$1^{c_1} 2^{c_2} \cdots n^{c_n}.$$

Since every element of $[n]$ belongs to exactly one cycle,

$$\boxed{\sum_{i=1}^{n} i c_i = n.}$$

Thus a cycle type is naturally an integer partition of n.

For example, if

$$\sigma = (1\,7\,5\,2)(3)(4\,12\,8)(6\,10)(9\,11) \in S_{12},$$

then σ has one 1-cycle, two 2-cycles, one 3-cycle, and one 4-cycle, so its cycle type is

$$1^1 2^2 3^1 4^1.$$

Proposition 2.2.2 (Permutations of a Fixed Cycle Type)**.** Fix nonnegative integers $c_1, \ldots, c_n$ satisfying $\sum_{i=1}^{n} i c_i = n$. The number of permutations in S_n having cycle type $1^{c_1} 2^{c_2} \cdots n^{c_n}$ is

$$\boxed{\frac{n!}{1^{c_1} c_1!\, 2^{c_2} c_2! \cdots n^{c_n} c_n!}.}$$

Proof. Start with an ordering of all n elements, giving $n!$ possibilities, and divide the ordered list into c_i blocks of size i for every i. Interpret each block as a cycle. A cycle of length i has i equivalent cyclic rotations, giving a factor i^{c_i} of overcounting, while the c_i cycles of the same length may be reordered in $c_i!$ ways without changing the permutation. Hence each permutation has been counted

$$\prod_{i=1}^{n} i^{c_i} c_i!$$

times, giving the stated formula. □

For $n = 4$, the possible cycle types give

cycle type	number
1^4	1
$1^2 2$	6
2^2	3
$1\,3$	8
4	6

$$1 + 6 + 3 + 8 + 6 = 24 = 4!.$$

Definition 2.2.3 (Cycle Indicator Polynomial)**.** Let $G \leq S_n$ be a subgroup. The *cycle indicator polynomial* of G is

$$\boxed{Z_G(t_1, \ldots, t_n) := \frac{1}{|G|} \sum_{\sigma \in G} \prod_{i=1}^{n} t_i^{c_i(\sigma)}.}$$

The exponent of t_i records the number of i-cycles in the permutation.

For example, the six permutations of S_3 consist of one permutation of cycle type 1^3, three of type $1\,2$, and two of type 3. Hence

$$\boxed{Z_{S_3}(t_1,t_2,t_3)=\frac{1}{3!}\left(t_1^3+3t_1t_2+2t_3\right).}$$

Using the fixed-cycle-type formula,

$$Z_{S_n}(t_1,\ldots,t_n)=\sum_{\substack{c_1,\ldots,c_n\geq 0\\ c_1+2c_2+\cdots+nc_n=n}}\frac{t_1^{c_1}\cdots t_n^{c_n}}{1^{c_1}c_1!\cdots n^{c_n}c_n!}.$$

Theorem 2.2.4 (Generating Function for Cycle Indicators)**.** The cycle indicator polynomials of the symmetric groups satisfy

$$\boxed{\sum_{n\geq 0}x^nZ_{S_n}(t_1,t_2,\ldots)=\exp\left(\sum_{j\geq 1}\frac{t_jx^j}{j}\right).}$$

Proof. For each $j\geq 1$,

$$\exp\left(\frac{t_jx^j}{j}\right)=\sum_{c_j\geq 0}\frac{t_j^{c_j}x^{jc_j}}{j^{c_j}c_j!}.$$

Therefore

$$\exp\left(\sum_{j\geq 1}\frac{t_jx^j}{j}\right)=\prod_{j\geq 1}\sum_{c_j\geq 0}\frac{t_j^{c_j}x^{jc_j}}{j^{c_j}c_j!}.$$

Expanding and collecting the coefficient of x^n gives

$$\sum_{\substack{c_1,c_2,\ldots\geq 0\\ \sum_{j\geq 1}jc_j=n}}\frac{t_1^{c_1}t_2^{c_2}\cdots}{1^{c_1}c_1!\,2^{c_2}c_2!\cdots}=Z_{S_n}(t_1,t_2,\ldots),$$

which proves the identity. □

2.3 Stirling Numbers

Two families of Stirling numbers connect set partitions, permutation cycles, and polynomial bases.

Recall that $S(n,k)$ denotes the Stirling number of the second kind, the number of set partitions of $[n]$ into exactly k nonempty blocks. We now introduce the corresponding numbers for permutation cycles.

Definition 2.3.1 (Unsigned Stirling Numbers of the First Kind)**.** Let $c(n,k)$ denote the number of permutations in S_n having exactly k cycles. These numbers are called the *unsigned Stirling numbers of the first kind*.

For example,

$$S(4,2)=7,\qquad c(4,2)=11.$$

To see the generating polynomial for $c(n,k)$, insert the element n into a permutation of $[n-1]$. Either n forms a new 1-cycle, increasing the number of cycles by one, or it is

inserted into one of the existing cycles. There are $n-1$ possible insertion positions in the latter case. Hence

$$\boxed{c(n,k) = c(n-1,k-1) + (n-1)c(n-1,k).}$$

Define the generating polynomial

$$C_n(t) := \sum_{k=0}^{n} c(n,k)t^k.$$

Using the recurrence,

$$\begin{aligned} C_n(t) &= \sum_{k=0}^{n} c(n,k)t^k = \sum_{k=0}^{n} \left[c(n-1,k-1) + (n-1)c(n-1,k)\right] t^k \\ &= \sum_{k=0}^{n} c(n-1,k-1)t^k + (n-1)\sum_{k=0}^{n} c(n-1,k)t^k. \end{aligned}$$

For the first sum, setting $j = k-1$ gives

$$\sum_{k=0}^{n} c(n-1,k-1)t^k = t\sum_{j=0}^{n-1} c(n-1,j)t^j = tC_{n-1}(t),$$

where $c(n-1,-1) = 0$. For the second sum, since $c(n-1,n) = 0$,

$$\sum_{k=0}^{n} c(n-1,k)t^k = C_{n-1}(t).$$

Therefore

$$\boxed{C_n(t) = (t+n-1)C_{n-1}(t).}$$

Starting from $C_0(t) = 1$ and iterating the recurrence,

$$\begin{aligned} C_n(t) &= (t+n-1)C_{n-1}(t) = (t+n-1)(t+n-2)C_{n-2}(t) \\ &= \cdots = t(t+1)(t+2)\cdots(t+n-1). \end{aligned}$$

Hence

$$\boxed{\sum_{k=0}^{n} c(n,k)t^k = t(t+1)(t+2)\cdots(t+n-1).}$$

Definition 2.3.2 (Falling Factorial)**.** For $k \geq 0$, the *falling factorial* is

$$(x)_k := x(x-1)(x-2)\cdots(x-k+1), \qquad (x)_0 := 1.$$

The two families

$$1, x, x^2, x^3, \ldots \qquad \text{and} \qquad 1, (x)_1, (x)_2, (x)_3, \ldots$$

are bases of the polynomial ring $\mathbb{C}[x]$.

Proposition 2.3.3 (Stirling Change of Basis)**.** For every $n \geq 0$,

$$\boxed{x^n = \sum_{k=0}^{n} S(n,k)(x)_k.}$$

Proof. First assume that x is a positive integer and consider all functions

$$f : [n] \longrightarrow [x], \qquad [n] = \{1, \ldots, n\}, \quad [x] = \{1, \ldots, x\}.$$

There are x^n such functions.
Fix $k = |\operatorname{Im}(f)|$. Write

$$\operatorname{Im}(f) = \{y_1, \ldots, y_k\},$$

where

$$\operatorname{Im}(f) := \{f(i) : i \in [n]\} \subseteq [x].$$

For each $y_j \in \operatorname{Im}(f)$, define its *fiber* (preimage) by

$$B_j := f^{-1}(y_j) = \{i \in [n] : f(i) = y_j\}.$$

Since $y_1, \ldots, y_k$ are exactly the values attained by f, the sets $B_1, \ldots, B_k$ are nonempty, pairwise disjoint, and satisfy

$$B_1 \cup \cdots \cup B_k = [n].$$

Hence

$$\{B_1, \ldots, B_k\}$$

is a set partition of $[n]$ into k nonempty blocks, giving $S(n, k)$ possibilities.
Conversely, fix a set partition

$$[n] = B_1 \sqcup \cdots \sqcup B_k.$$

To construct a function whose fibers are exactly these k blocks, assign distinct values $y_1, \ldots, y_k \in [x]$ to the blocks and define

$$f(i) = y_j \qquad \text{for } i \in B_j.$$

The number of injective assignments

$$\{B_1, \ldots, B_k\} \longrightarrow [x]$$

is

$$(x)_k = x(x-1)\cdots(x-k+1).$$

Therefore

$$\#\{f : [n] \to [x] : |\operatorname{Im}(f)| = k\} = S(n, k)(x)_k.$$

Every function $f : [n] \to [x]$ has a unique value of $k = |\operatorname{Im}(f)|$. Summing over k gives

$$x^n = \sum_{k=0}^{n} S(n, k)(x)_k.$$

This identity has been proved for every positive integer x. Since both sides are polynomials in x and agree at infinitely many values of x, they are identical as polynomials. Hence

$$\boxed{x^n = \sum_{k=0}^{n} S(n, k)(x)_k.}$$

□

Define the *signed Stirling number of the first kind* by

$$s(n,k) := (-1)^{n-k}c(n,k).$$

Proposition 2.3.4. The inverse change of basis is

$$\boxed{(x)_n = \sum_{k=0}^{n} s(n,k)x^k = \sum_{k=0}^{n}(-1)^{n-k}c(n,k)x^k.}$$

Consequently, the infinite lower-triangular matrices

$$\Big(S(n,k)\Big)_{n,k\geq 0} \qquad \text{and} \qquad \Big(s(n,k)\Big)_{n,k\geq 0}$$

are inverse change-of-basis matrices.

Proof. Since

$$(x)_n = x(x-1)\cdots(x-n+1),$$

replacing t by $-x$ in

$$\sum_{k=0}^{n} c(n,k)t^k = t(t+1)\cdots(t+n-1)$$

and multiplying by $(-1)^n$ gives

$$(x)_n = \sum_{k=0}^{n}(-1)^{n-k}c(n,k)x^k.$$

Together with $x^n = \sum_k S(n,k)(x)_k$, these are inverse changes between the two polynomial bases. □

2.4 Exponential Formula

The exponential formula describes labeled structures that decompose uniquely into unordered connected components.
Let $\mathcal{C}$ be a combinatorial class of connected labeled structures, and let

$$C(x) = \sum_{n\geq 1} c_n \frac{x^n}{n!},$$

where c_n is the number of $\mathcal{C}$-objects on the labeled set $[n]$.
Let $\mathcal{A}$ be a combinatorial class whose objects are unordered sets of disjoint $\mathcal{C}$-objects, and let

$$A(x) = \sum_{n\geq 0} a_n \frac{x^n}{n!}.$$

Theorem 2.4.1 (Exponential Formula)**.** If every $\mathcal{A}$-object decomposes uniquely into an unordered set of connected $\mathcal{C}$-objects, then

$$\boxed{A(x) = \exp(C(x)).}$$

Proof. Let $a_{n,k}$ denote the number of $\mathcal{A}$-objects on the labeled set $[n]$ that decompose into exactly k $\mathcal{C}$-objects, and define

$$A_k(x) := \sum_{n\geq 0} a_{n,k}\frac{x^n}{n!}.$$

Fix k component sizes

$$j_1,\dots,j_k \geq 1, \qquad j_1+\cdots+j_k = n.$$

Temporarily order the k components as $1,\dots,k$. To choose which labels belong to each component, partition $[n]$ into ordered subsets

$$B_1 \sqcup \cdots \sqcup B_k = [n], \qquad |B_r| = j_r.$$

The number of such choices is the multinomial coefficient

$$\binom{n}{j_1,\dots,j_k} = \frac{n!}{j_1!\cdots j_k!}.$$

For each label set B_r of size j_r, there are c_{j_r} possible $\mathcal{C}$-objects on B_r. Since the components are chosen independently, the number of ordered collections of components with these sizes is

$$\frac{n!}{j_1!\cdots j_k!}\, c_{j_1}\cdots c_{j_k}.$$

The components of an $\mathcal{A}$-object form an unordered set. Thus each unordered collection has been counted $k!$ times by temporarily ordering its components. Therefore

$$a_{n,k} = \frac{1}{k!}\sum_{\substack{j_1+\cdots+j_k=n\\ j_1,\dots,j_k\geq 1}} \frac{n!}{j_1!\cdots j_k!}\, c_{j_1}\cdots c_{j_k}.$$

Now substitute this expression into the EGF:

$$A_k(x) = \sum_{n\geq 0} a_{n,k}\frac{x^n}{n!} = \frac{1}{k!}\sum_{j_1,\dots,j_k\geq 1}\frac{c_{j_1}\cdots c_{j_k}}{j_1!\cdots j_k!}x^{j_1+\cdots+j_k} = \frac{1}{k!}\prod_{r=1}^{k}\left(\sum_{j_r\geq 1} c_{j_r}\frac{x^{j_r}}{j_r!}\right) = \frac{C(x)^k}{k!}.$$

Finally, every $\mathcal{A}$-object has a unique number $k \geq 0$ of components, so

$$A(x) = \sum_{k\geq 0} A_k(x) = \sum_{k\geq 0}\frac{C(x)^k}{k!} = \boxed{\exp(C(x))}.$$

□

Example 2.4.2 (Permutations as Sets of Cycles)**.** A permutation decomposes uniquely into an unordered set of disjoint cycles. Thus the connected components are cycles.
For $n \geq 1$, the number of cycles on a fixed set of n labeled elements is $(n-1)!$. Hence the EGF for cycles is

$$C(x) = \sum_{n\geq 1}(n-1)!\frac{x^n}{n!} = \sum_{n\geq 1}\frac{x^n}{n} = -\log(1-x).$$

By the exponential formula, the EGF for all permutations is

$$A(x) = \exp(C(x)) = \exp(-\log(1-x)) = \boxed{\frac{1}{1-x}}.$$

This agrees with the direct computation

$$A(x) = \sum_{n\geq 0} n!\frac{x^n}{n!} = \sum_{n\geq 0} x^n = \frac{1}{1-x}.$$

Chapter 3

Labeled Trees and Lagrange Inversion

3.1 Labeled Trees

A *labeled tree on* $[n]$ is a tree whose vertex set is $[n] = \{1, 2, \ldots, n\}$. Let t_n denote the number of labeled trees on $[n]$. The first few values suggest

$$t_1 = 1, \qquad t_2 = 1, \qquad t_3 = 3, \qquad t_4 = 16, \qquad t_5 = 125,$$

leading to the formula

$$\boxed{t_n = n^{n-2}.}$$

This is *Cayley's formula*; we will derive it using exponential generating functions and Lagrange inversion.

Definition 3.1.1 (Rooted Labeled Tree)**.** A *rooted labeled tree* is a labeled tree together with a distinguished vertex called the *root*. Let r_n denote the number of rooted labeled trees on $[n]$.

Since any of the n vertices of a labeled tree may be chosen as the root,

$$\boxed{r_n = nt_n.}$$

Example 3.1.2 ($n = 4$)**.** Fix the labeled tree with edges

$$\{1, 2\}, \qquad \{2, 3\}, \qquad \{3, 4\}.$$

Each of its four vertices can be chosen as the root, giving four different rooted labeled trees.

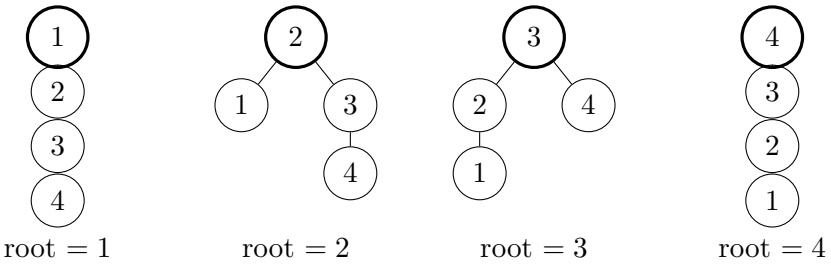


Figure 2: The same labeled tree on [4] with each of its four vertices chosen as the root.

Thus each labeled tree on [4] gives exactly 4 rooted labeled trees. In general, each labeled tree on $[n]$ gives exactly n rooted labeled trees, so

$$\boxed{r_n = nt_n.}$$

Define the EGF for rooted labeled trees by

$$R(x) := \sum_{n \geq 1} r_n \frac{x^n}{n!}.$$

Definition 3.1.3 (Rooted Forest)**.** A *rooted forest* is an unordered set of rooted labeled trees on disjoint sets of labels. Let f_n denote the number of rooted forests on $[n]$, with $f_0 = 1$, and define

$$F(x) := \sum_{n \geq 0} f_n \frac{x^n}{n!}.$$

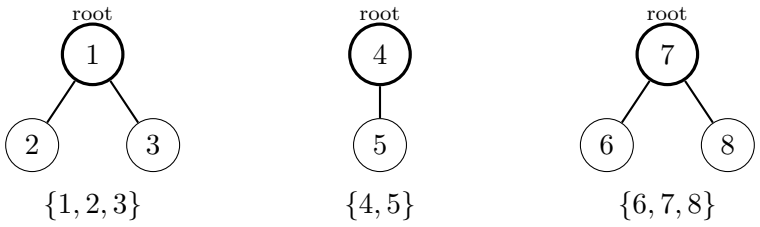


Figure 3: A rooted forest on [8] with three rooted-tree components.

A rooted forest is an unordered set of rooted trees. Therefore, by the exponential formula,

$$\boxed{F(x) = \exp(R(x)).}$$

We now relate rooted trees to rooted forests directly. Given a rooted forest on $n-1$ labeled vertices, introduce one new vertex and connect it to the root of every tree in the forest. Declare the new vertex to be the root. This produces a rooted tree on n vertices. Conversely, deleting the root of a rooted tree leaves a rooted forest whose components are rooted at the former neighbors of the deleted root. Hence a rooted tree consists combinatorially of one root together with an unordered set of rooted trees.
In terms of EGFs, the distinguished root contributes a factor x, while the unordered set of rooted-tree components contributes $F(x) = e^{R(x)}$. Therefore

$$\boxed{R(x) = xF(x) = xe^{R(x)}.}$$

This functional equation determines the rooted-tree EGF implicitly. Equivalently,

$$R(x)e^{-R(x)} = x.$$

To extract the coefficients of $R(x)$, we use compositional inverses and Lagrange inversion.

3.2 Lagrange Inversion

Definition 3.2.1 (Compositional Inverse)**.** Let $A(x) = a_1x + a_2x^2 + \cdots \in R[[x]]$ have zero constant term. A formal power series $B(x)$ is called a *compositional inverse* of $A(x)$ if

$$A(B(x)) = x \qquad \text{and} \qquad B(A(x)) = x.$$

When it exists, it is denoted by

$$B(x) = A^{\langle -1 \rangle}(x).$$

Proposition 3.2.2. If $A(x) = a_1x + a_2x^2 + \cdots \in R[[x]]$ and a_1 is a unit in R, then $A(x)$ has a unique compositional inverse in $R[[x]]$.

The condition on a_1 is analogous to the condition on the constant coefficient for multiplicative inverses. Indeed, if

$$B(x) = b_1x + b_2x^2 + \cdots$$

satisfies $A(B(x)) = x$, then comparison of coefficients first gives $a_1b_1 = 1$, and the higher coefficients $b_2, b_3, \ldots$ are then determined recursively.

For the rooted-tree equation, define

$$A(u) := ue^{-u}.$$

Since

$$R(x)e^{-R(x)} = x,$$

we have

$$A(R(x)) = x.$$

Thus

$$\boxed{R(x) = A^{\langle -1\rangle}(x), \qquad A(x) = xe^{-x}.}$$

Theorem 3.2.3 (Lagrange Inversion Theorem)**.** Let $A(x) = x/\phi(x) \in R[[x]]$, where $\phi(0)$ is a unit, and let $B(x) = A^{\langle -1\rangle}(x)$. Then, for $n \geq 1$,

$$\boxed{[x^n]B(x) = \frac{1}{n}[u^{n-1}]\phi(u)^n.}$$

For rooted trees,

$$R(x) = xe^{R(x)},$$

so $\phi(u) = e^u$. Lagrange inversion gives

$$[x^n]R(x) = \frac{1}{n}[u^{n-1}]e^{nu}.$$

Since

$$e^{nu} = \sum_{m\geq 0} \frac{n^m}{m!}u^m,$$

we have

$$[u^{n-1}]e^{nu} = \frac{n^{n-1}}{(n-1)!}.$$

Therefore

$$[x^n]R(x) = \frac{1}{n}\frac{n^{n-1}}{(n-1)!} = \boxed{\frac{n^{n-1}}{n!}}.$$

On the other hand, by the definition of the EGF,

$$R(x) = \sum_{n\geq 1} r_n \frac{x^n}{n!},$$

so

$$[x^n]R(x) = \frac{r_n}{n!}.$$

Comparing the two expressions gives

$$\boxed{r_n = n^{n-1}.}$$

Finally, since $r_n = nt_n$,

$$t_n = \frac{r_n}{n} = n^{n-2}.$$

Theorem 3.2.4 (Cayley's Formula)**.** The number of labeled trees on $[n]$ is

$$\boxed{t_n = n^{n-2}.}$$

Equivalently, the number of rooted labeled trees on $[n]$ is

$$\boxed{r_n = n^{n-1}.}$$

Chapter 4

q-Analogs

4.1 q-Integers and Gaussian Binomial Coefficients

The basic idea of a q-analog is to replace an integer-valued combinatorial quantity by a polynomial or rational expression in an indeterminate q, which recovers the original quantity when $q \to 1$.

Definition 4.1.1 (q-Integer)**.** For $n \geq 1$, the *q-integer* $[n]_q$ is defined by

$$\boxed{[n]_q := 1 + q + q^2 + \cdots + q^{n-1} = \frac{1-q^n}{1-q}.}$$

In particular,

$$[n]_q\Big|_{q=1} = n.$$

Definition 4.1.2 (q-Factorial)**.** For $n \geq 1$, define

$$[n]_q! := [n]_q[n-1]_q \cdots [2]_q[1]_q, \qquad [0]_q! := 1.$$

The q-analog of the ordinary binomial coefficient is the Gaussian binomial coefficient.

Definition 4.1.3 (Gaussian Binomial Coefficient)**.** For $0 \leq k \leq n$, define

$$\boxed{\binom{n}{k}_q := \frac{[n]_q!}{[k]_q![n-k]_q!}.}$$

It is also called the *q-binomial coefficient*.

Although this expression is initially written as a quotient, $\binom{n}{k}_q$ is in fact a polynomial in q with nonnegative integer coefficients.
Since $[m]_q \to m$ as $q \to 1$,

$$\boxed{\binom{n}{k}_q\Big|_{q=1} = \binom{n}{k}.}$$

The factorial formula immediately gives the symmetry

$$\boxed{\binom{n}{k}_q = \binom{n}{n-k}_q.}$$

Proposition 4.1.4 (q-Pascal Identities)**.** For $1 \leq k \leq n-1$,

$$\boxed{\binom{n}{k}_q = \binom{n-1}{k-1}_q + q^k\binom{n-1}{k}_q,}$$

and equivalently,

$$\boxed{\binom{n}{k}_q = q^{n-k}\binom{n-1}{k-1}_q + \binom{n-1}{k}_q.}$$

These are the two q-analogs of Pascal's identity. They are equivalent under the symmetry $k \leftrightarrow n-k$.

Definition 4.1.5 (Partitions in a Rectangle)**.** A partition $\lambda = (\lambda_1, \ldots, \lambda_k)$ *fits inside a* $k \times (n-k)$ *rectangle* if

$$n - k \geq \lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_k \geq 0.$$

Its size is

$$|\lambda| := \lambda_1 + \cdots + \lambda_k,$$

the number of boxes in its Ferrers diagram.

A lattice path along the boundary of a Ferrers diagram inside a $k \times (n-k)$ rectangle determines the partition uniquely. The number of boxes on one side of the path is $|\lambda|$, so weighting a partition λ by $q^{|\lambda|}$ gives the generating polynomial

$$\boxed{\binom{n}{k}_q = \sum_{\lambda \subseteq k \times (n-k)} q^{|\lambda|}.}$$

Thus the coefficient of q^m in $\binom{n}{k}_q$ counts the partitions of m whose Ferrers diagrams fit inside a $k \times (n-k)$ rectangle:

$$\boxed{[q^m]\binom{n}{k}_q = \#\left\{\lambda \subseteq k \times (n-k) : |\lambda| = m\right\}.}$$

Example 4.1.6. For $n = 4$ and $k = 2$, the partitions fitting inside a 2×2 rectangle have sizes

$$0,\ 1,\ 2,\ 2,\ 3,\ 4.$$

Therefore

$$\boxed{\binom{4}{2}_q = 1 + q + 2q^2 + q^3 + q^4.}$$

Setting $q = 1$ gives $1 + 1 + 2 + 1 + 1 = 6 = \binom{4}{2}$.

The symmetry

$$\binom{n}{k}_q = \binom{n}{n-k}_q$$

also has a geometric interpretation: conjugating the Ferrers diagram exchanges partitions inside a $k \times (n-k)$ rectangle with partitions inside an $(n-k) \times k$ rectangle.

4.2 Finite-Field Interpretation

We now give another combinatorial interpretation of the Gaussian binomial coefficient.

Definition 4.2.1 (Finite Field)**.** Let $q = p^m$, where p is prime and $m \geq 1$. The notation $\mathbb{F}_q$ denotes a finite field with exactly q elements:

$$|\mathbb{F}_q| = q.$$

The vector space $\mathbb{F}_q^n$ consists of all n-tuples with entries in $\mathbb{F}_q$.

An r-dimensional vector space over $\mathbb{F}_q$ contains exactly q^r vectors, since every vector is uniquely determined by its r coordinates relative to a basis.

Theorem 4.2.2 (Finite-Field Interpretation)**.** For $0 \le k \le n$,

$$\boxed{\binom{n}{k}_q = \#\left\{k\text{-dimensional subspaces of } \mathbb{F}_q^n\right\}.}$$

We prove this using reduced row echelon form.

Definition 4.2.3 (Reduced Row Echelon Form)**.** A matrix is in *reduced row echelon form (RREF)* if:

1. all zero rows occur below all nonzero rows;
2. the first nonzero entry of every nonzero row is 1, called its *pivot*;
3. the pivot of each lower row occurs strictly to the right of the pivot of the row above it;
4. every pivot is the only nonzero entry in its column.

A column containing a pivot is called a *pivot column*; all other columns are called *free columns*.

Lemma 4.2.4. If two matrices A and B are both in RREF and have the same row space, then

$$A = B.$$

Thus every k-dimensional subspace $V \le \mathbb{F}_q^n$ is represented uniquely as the row space of a $k \times n$ full-rank matrix in RREF. Therefore there is a bijection

$$\left\{k\text{-dimensional subspaces of } \mathbb{F}_q^n\right\} \longleftrightarrow \left\{k \times n \text{ full-rank RREF matrices over } \mathbb{F}_q\right\}.$$

Let $\mathcal{S}_1$ denote the set of k-dimensional subspaces of $\mathbb{F}_q^n$, and let $\mathcal{S}_2$ denote the set of $k \times n$ full-rank RREF matrices. The bijection is

$$\pi_1 : \mathcal{S}_2 \longrightarrow \mathcal{S}_1, \qquad A \longmapsto \mathrm{Row}(A).$$

We next associate an RREF matrix with a partition. A full-rank $k \times n$ RREF matrix has exactly k pivot columns and $n - k$ free columns. For each row, count the number of free positions in which an arbitrary field element may occur. Reading these numbers from bottom to top gives a weakly decreasing sequence

$$\lambda = (\lambda_1, \ldots, \lambda_k), \qquad n - k \ge \lambda_1 \ge \cdots \ge \lambda_k \ge 0.$$

Hence λ is a partition fitting inside a $k \times (n-k)$ rectangle.
This defines a map

$$\pi_2 : \mathcal{S}_2 \longrightarrow \left\{\lambda : \lambda \subseteq k \times (n-k)\right\}.$$

For a fixed partition λ, the pivot positions are determined, while the $|\lambda|$ free entries may be chosen independently from $\mathbb{F}_q$. Each free entry has q possible values, so

$$\boxed{|\pi_2^{-1}(\lambda)| = q^{|\lambda|}.}$$

Therefore

$$|\mathcal{S}_1| = |\mathcal{S}_2| = \sum_{\lambda \subseteq k \times (n-k)} q^{|\lambda|}.$$

By the partition interpretation of the Gaussian binomial coefficient,

$$\sum_{\lambda \subseteq k \times (n-k)} q^{|\lambda|} = \binom{n}{k}_q .$$

Hence

$$\boxed{\#\left\{k\text{-dimensional subspaces of } \mathbb{F}_q^n\right\} = \binom{n}{k}_q .}$$

There is also a direct counting formula. An ordered linearly independent k-tuple $(v_1, \ldots, v_k)$ in $\mathbb{F}_q^n$ may be chosen in

$$(q^n - 1)(q^n - q) \cdots (q^n - q^{k-1})$$

ways: after choosing $v_1, \ldots, v_{r-1}$, their span contains q^{r-1} vectors, so v_r has $q^n - q^{r-1}$ choices. Every fixed k-dimensional subspace has

$$(q^k - 1)(q^k - q) \cdots (q^k - q^{k-1})$$

ordered bases. Dividing gives

$$\boxed{\binom{n}{k}_q = \frac{(q^n - 1)(q^n - q) \cdots (q^n - q^{k-1})}{(q^k - 1)(q^k - q) \cdots (q^k - q^{k-1})}.}$$

Equivalently, cancelling the powers of q,

$$\boxed{\binom{n}{k}_q = \prod_{r=0}^{k-1} \frac{q^{n-r} - 1}{q^{k-r} - 1}.}$$

Chapter 5

Posets and Lattices

5.1 Partially Ordered Sets

Definition 5.1.1 (Partially Ordered Set)**.** A *partially ordered set*, or *poset*, is a pair $(P, \leq)$, where P is a set and $\leq$ is a binary relation satisfying, for all $x, y, z \in P$:

1. **Reflexivity:** $x \leq x$.
2. **Antisymmetry:** $x \leq y$ and $y \leq x$ imply $x = y$.
3. **Transitivity:** $x \leq y$ and $y \leq z$ imply $x \leq z$.

Two elements $x, y \in P$ are *comparable* if $x \leq y$ or $y \leq x$; otherwise, they are *incomparable*. We write $x < y$ if $x \leq y$ and $x \neq y$.

Example 5.1.2 (A Simple Poset)**.** Consider the poset $P = \{a, b, c, d\}$ with $a < b < c$ and $a < b < d$, where c and d are incomparable.

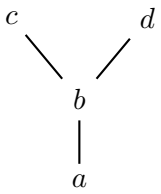


Figure 4: A Hasse diagram of a poset with incomparable elements c and d.

Although $a < b < c$ and $a < b < d$, neither $c \leq d$ nor $d \leq c$ holds. Thus P is not totally ordered.

Definition 5.1.3 (Covering Relation)**.** For $x, y \in P$, we say that y *covers* x, written $x \lessdot y$, if $x < y$ and there is no $z \in P$ such that $x < z < y$.

Remark 5.1.4. In Figure 4, although $a < c$, the element c does not cover a, since $a < b < c$. Instead, b covers a, and both c and d cover b.

A *Hasse diagram* represents a finite poset by placing larger elements above smaller ones and drawing an edge between x and y precisely when $x \lessdot y$. Reflexive and transitive relations are omitted.

Definition 5.1.5 (Chains and Antichains)**.** A *chain* is a subset of P in which every two elements are comparable. An *antichain* is a subset of P in which every two distinct elements are incomparable.

Example 5.1.6 (Chains and Antichains)**.** The following diagrams illustrate a chain, an antichain, and a poset that is neither.

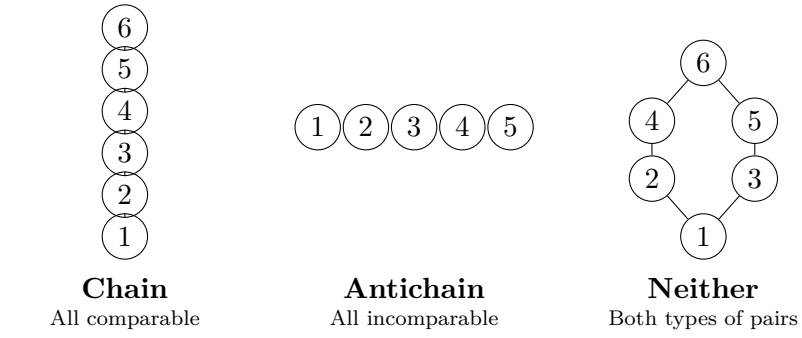


Figure 5: Examples of a chain, an antichain, and a poset that is neither.

In (a), every pair of elements is comparable. In (b), every pair of distinct elements is incomparable. In (c), $1 < 2$, while 2 and 3 are incomparable.

A chain may be written as $x_0 < x_1 < \cdots < x_r$, and its *length* is r, the number of strict steps. A chain is *maximal* if it is not properly contained in another chain. It is *saturated* if $x_{i-1} \lessdot x_i$ for every $1 \le i \le r$.

Remark 5.1.7. A maximal chain is necessarily saturated, but a saturated chain need not be maximal. The former cannot be extended, while the latter only requires that no element can be inserted between consecutive elements.

A poset is *totally ordered* if every two elements are comparable.

Definition 5.1.8 (Poset Isomorphism)**.** Two posets $(P, \le_P)$ and $(Q, \le_Q)$ are *isomorphic* if there exists a bijection $\phi : P \to Q$ such that

$$\boxed{x \le_P y \iff \phi(x) \le_Q \phi(y).}$$

Such a map ϕ is called a *poset isomorphism*.

Example 5.1.9 (Prime Factorization and Poset Isomorphism)**.** Every positive integer n has a unique prime factorization

$$n = 2^{a_1} 3^{a_2} 5^{a_3} 7^{a_4} \cdots,$$

where $a_i \ge 0$ and only finitely many a_i are nonzero. Define the *exponent vector* by $\phi(n) := (a_1, a_2, a_3, \ldots)$. For example,

$$12 = 2^2 \cdot 3 \longmapsto (2, 1, 0, \ldots), \qquad 20 = 2^2 \cdot 5 \longmapsto (2, 0, 1, \ldots).$$

Let $\mathcal{E}$ be the set of all nonnegative integer sequences with finitely many nonzero entries, ordered componentwise:

$$(a_i)_{i \ge 1} \le (b_i)_{i \ge 1} \iff a_i \le b_i \quad \text{for every } i.$$

By uniqueness of prime factorization, ϕ is a bijection from the positive integers to $\mathcal{E}$. Moreover,

$$\boxed{a \mid b \iff \phi(a) \le \phi(b).}$$

Hence the positive integers ordered by divisibility are isomorphic to $\mathcal{E}$.
In particular, consider the divisor poset

$$D(20) = \{1, 2, 4, 5, 10, 20\}.$$

Since $20 = 2^2 \cdot 5$, its exponent vectors are

$$\begin{aligned} 1 &\longmapsto (0,0), \quad 2 \longmapsto (1,0), \quad 4 \longmapsto (2,0), \\ 5 &\longmapsto (0,1), \quad 10 \longmapsto (1,1), \quad 20 \longmapsto (2,1). \end{aligned}$$

Therefore

$$\boxed{D(20) \cong \{0,1,2\} \times \{0,1\},}$$

where the right-hand side has the componentwise order.

5.2 Important Examples of Posets

Example 5.2.1 (Boolean Poset)**.** The *Boolean poset* B_n consists of all subsets of $[n]$, ordered by inclusion:

$$B_n := (2^{[n]}, \subseteq), \qquad S \leq T \iff S \subseteq T.$$

Here $2^{[n]}$ denotes the power set of $[n]$. In particular, $|B_n| = 2^n$, and T covers S precisely when $S \subseteq T$ and $|T| = |S| + 1$.

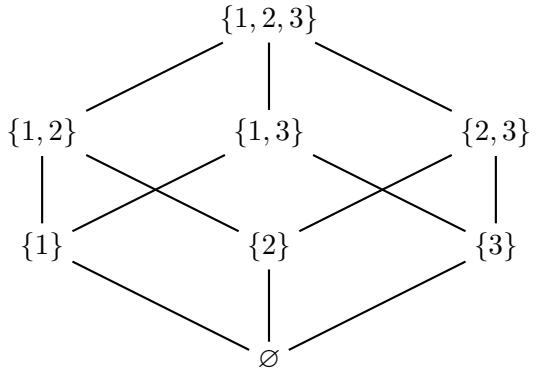


Figure 6: Hasse diagram of the Boolean poset B_3.

Example 5.2.2 (Divisor Poset)**.** For a positive integer m, let $D(m)$ denote the set of all positive divisors of m, ordered by divisibility:

$$a \leq b \iff a \mid b.$$

For example, $D(6) = \{1, 2, 3, 6\}$, with covering relations $1 \lessdot 2$, $1 \lessdot 3$, $2 \lessdot 6$, and $3 \lessdot 6$.

Example 5.2.3 (Young's Lattice)**.** Let Y denote the set of all integer partitions. We write $\mu = (\mu_1, \mu_2, \ldots)$ and $\lambda = (\lambda_1, \lambda_2, \ldots)$, appending zeros when necessary. The order relation is

$$\boxed{\mu \leq \lambda \iff \mu_i \leq \lambda_i \quad \text{for every } i \geq 1.}$$

For $n \geq 0$, let Y_n denote the subposet consisting of partitions of size at most n, where $|\lambda| := \sum_i \lambda_i$.

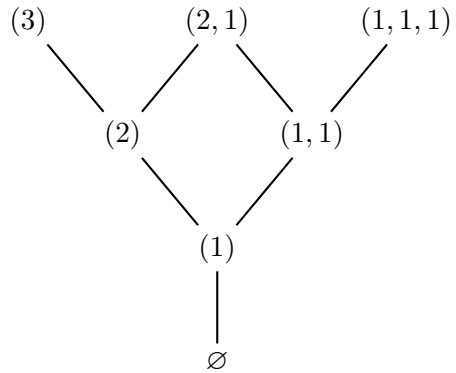


Figure 7: Hasse diagram of Young's lattice Y_3, consisting of all integer partitions of size at most 3.

Example 5.2.4 (Set Partition Lattice)**.** Let Π_n denote the set of all set partitions of $[n]$. For $\pi, \sigma \in \Pi_n$, define the *refinement order* by

$$\boxed{\pi \leq \sigma \iff \text{every block of } \pi \text{ is contained in some block of } \sigma.}$$

Thus $\pi \leq \sigma$ means that π is finer than σ.

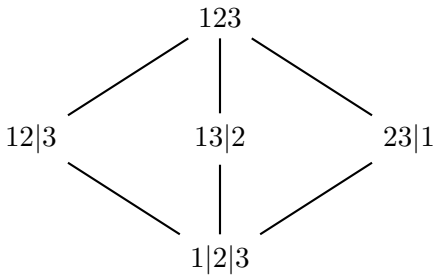


Figure 8: Hasse diagram of the set partition lattice Π_3 under refinement order.

Example 5.2.5 (Dominance Order)**.** Let $\mathcal{D}(n)$ denote the set of all integer partitions of n. For $\mu, \lambda \in \mathcal{D}(n)$, define the *dominance order* by

$$\boxed{\mu \leq_{\text{dom}} \lambda \iff \sum_{i=1}^{j} \mu_i \leq \sum_{i=1}^{j} \lambda_i \quad \text{for every } j \geq 1.}$$

For example, under the dominance order,

$$(4,3) \leq_{\text{dom}} (5,2), \qquad (2,2,1,1) \leq_{\text{dom}} (3,2,1).$$

Indeed, comparing partial sums gives

$$\begin{aligned} (4,3) \leq_{\text{dom}} (5,2) &\iff 4 \leq 5, \quad 4+3 = 5+2, \\ (2,2,1,1) \leq_{\text{dom}} (3,2,1) &\iff 2 \leq 3, \quad 4 \leq 5, \quad 5 \leq 6, \quad 6 = 6. \end{aligned}$$

However, $(4,1,1)$ and $(3,3)$ are incomparable, since

$$4 > 3, \qquad 4+1 < 3+3.$$

Thus neither $(4,1,1) \leq_{\text{dom}} (3,3)$ nor $(3,3) \leq_{\text{dom}} (4,1,1)$ holds.

Remark 5.2.6. Young's lattice and the dominance order are different partial orders on integer partitions. Young's lattice compares individual parts, while the dominance order compares partial sums.

Proposition 5.2.7. The dominance order on $\mathcal{D}(n)$ is a total order for $n \leq 5$, but not for $n \geq 6$.

Proof. For $n \leq 5$, the partitions can be checked directly and form a chain under dominance. For $n = 6$, consider $\mu = (4,1,1)$ and $\lambda = (3,3)$. Their first partial sums satisfy $4 > 3$, while their first two partial sums satisfy $4+1 < 3+3$. Thus neither $\mu \leq_{\text{dom}} \lambda$ nor $\lambda \leq_{\text{dom}} \mu$. For $n > 6$, appending a common first part $n-6$ when it is sufficiently large, or equivalently extending the two partitions by a common partition, preserves incomparability. □

5.3 Graded Posets and Rank Generating Functions

Definition 5.3.1 (Graded Poset)**.** A finite poset P is *graded* if all maximal chains have the same length. This common length is called the *rank* of P, denoted by $\operatorname{rank}(P)$.

Definition 5.3.2 (Rank Function)**.** Let P be a finite graded poset. Its *rank function* is the map $\rho : P \to \mathbb{Z}_{\geq 0}$ defined by the length of a maximal chain from a minimal element to x. Equivalently, ρ is characterized by:

1. $\rho(x) = 0$ for every minimal element $x \in P$;
2. $\rho(y) = \rho(x) + 1$ whenever $x \lessdot y$.

For $0 \leq i \leq \operatorname{rank}(P)$, define the *$i$-th rank level* by $P_i := \{x \in P : \rho(x) = i\}$.

Definition 5.3.3 (Rank Generating Function)**.** The *rank generating function* of a finite graded poset P is

$$\boxed{F(P, z) := \sum_{x \in P} z^{\rho(x)} = \sum_{i=0}^{\operatorname{rank}(P)} |P_i| z^i.}$$

Example 5.3.4 (Boolean Poset)**.** In B_n, the rank function is $\rho(S) = |S|$. Since there are $\binom{n}{k}$ subsets of size k,

$$\boxed{F(B_n, z) = \sum_{k=0}^{n} \binom{n}{k} z^k = (1 + z)^n.}$$

In particular, $F(B_2, z) = 1 + 2z + z^2$.

Definition 5.3.5 (Interval)**.** For $x \leq y$ in a poset P, the *closed interval* from x to y is

$$[x, y] := \{z \in P : x \leq z \leq y\}.$$

Example 5.3.6. In the divisor poset $D(20)$,

$$[1, 10] = \{1, 2, 5, 10\}.$$

5.4 Operations on Posets

Let $(P, \leq_P)$ and $(Q, \leq_Q)$ be two posets.

Definition 5.4.1 (Disjoint Union of Posets)**.** The *disjoint union* $P \sqcup Q$ consists of all elements of P and Q, with the original order relations preserved within each poset and no comparability between elements from different posets.

Definition 5.4.2 (Cartesian Product of Posets)**.** The *Cartesian product* $P \times Q$ is the set of ordered pairs (p, q) with $p \in P$ and $q \in Q$, equipped with the *product order*

$$\boxed{(p_1, q_1) \leq (p_2, q_2) \iff p_1 \leq_P p_2 \text{ and } q_1 \leq_Q q_2.}$$

Lemma 5.4.3. A covering relation in $P \times Q$ has exactly one of the following forms:

$$(p, q) \lessdot (p', q) \quad \text{with } p \lessdot_P p',$$

or

$$(p, q) \lessdot (p, q') \quad \text{with } q \lessdot_Q q'.$$

Proof. If both coordinates increase strictly, then

$$(p, q) < (p', q) < (p', q'),$$

so the relation cannot be a covering relation. Thus exactly one coordinate changes. If that coordinate does not form a covering relation in its original poset, an intermediate element can also be inserted in the product. The converse follows directly from the definition of the product order. □

Proposition 5.4.4 (Rank of Cartesian Products)**.** Suppose P and Q are finite graded posets of ranks r_P and r_Q, respectively. Then $P \times Q$ is graded, with

$$\boxed{\operatorname{rank}(P \times Q) = r_P + r_Q.}$$

Moreover,

$$\boxed{F(P \times Q, z) = F(P, z)F(Q, z).}$$

Proof. Let ρ_P and ρ_Q denote the rank functions of P and Q. Define

$$\rho(p, q) := \rho_P(p) + \rho_Q(q).$$

If (p, q) is minimal in $P \times Q$, then p and q are minimal in their respective posets, so $\rho(p, q) = 0$.
By the preceding lemma, a covering relation in $P \times Q$ increases exactly one coordinate by a covering relation. Hence, whenever $(p, q) \lessdot (p', q')$,

$$\rho(p', q') = \rho(p, q) + 1.$$

Thus ρ is a rank function, and its maximum value is $r_P + r_Q$. Therefore $P \times Q$ is graded of rank $r_P + r_Q$.
For the rank generating function,

$$\begin{aligned} F(P \times Q, z) &= \sum_{(p,q) \in P \times Q} z^{\rho(p,q)} = \sum_{p \in P} \sum_{q \in Q} z^{\rho_P(p) + \rho_Q(q)} \\ &= \left(\sum_{p \in P} z^{\rho_P(p)} \right) \left(\sum_{q \in Q} z^{\rho_Q(q)} \right) = F(P, z)F(Q, z). \end{aligned}$$

□

Example 5.4.5 (Products of Chains)**.** Let $[m]$ denote the chain $1 < 2 < \cdots < m$. Its rank generating function is

$$F([m], z) = 1 + z + \cdots + z^{m-1} = [m]_z.$$

Therefore

$$\boxed{F([m_1] \times \cdots \times [m_k], z) = \prod_{i=1}^{k} [m_i]_z.}$$

5.5 Meets, Joins, and Lattices

Definition 5.5.1 (Meet)**.** For $x, y \in P$, a *meet* of x and y, denoted by $x \wedge y$, is their greatest lower bound. That is, $m = x \wedge y$ if:

1. $m \leq x$ and $m \leq y$;
2. for every $z \in P$ with $z \leq x$ and $z \leq y$, we have $z \leq m$.

Definition 5.5.2 (Join)**.** For $x, y \in P$, a *join* of x and y, denoted by $x \vee y$, is their least upper bound. That is, $j = x \vee y$ if:

1. $x \leq j$ and $y \leq j$;
2. for every $z \in P$ with $x \leq z$ and $y \leq z$, we have $j \leq z$.

Proposition 5.5.3 (Basic Properties of Meet and Join)**.** Whenever the relevant meets and joins exist, they satisfy:

1. **Commutativity:**
$$x \wedge y = y \wedge x, \qquad x \vee y = y \vee x.$$
2. **Associativity:**
$$(x \wedge y) \wedge z = x \wedge (y \wedge z), \qquad (x \vee y) \vee z = x \vee (y \vee z).$$
3. **Order characterization:**
$$x \wedge y = x \iff x \leq y, \qquad x \vee y = x \iff y \leq x.$$
4. **Uniqueness:** If $x \wedge y$ or $x \vee y$ exists, then it is unique.

Proof. Commutativity follows directly from the symmetric definitions of meet and join. Associativity follows because both sides are the greatest lower bound or least upper bound of the same three elements. The order characterization follows directly from the definitions.
For uniqueness, suppose m and m' are both meets of x and y. Since m' is a lower bound and m is the greatest lower bound, $m' \leq m$. Similarly, $m \leq m'$. By antisymmetry, $m = m'$. The proof for joins is analogous. □

Definition 5.5.4 (Semilattices and Lattices)**.** A poset P is called:

1. a *meet-semilattice* if $x \wedge y$ exists for every $x, y \in P$;
2. a *join-semilattice* if $x \vee y$ exists for every $x, y \in P$;
3. a *lattice* if both $x \wedge y$ and $x \vee y$ exist for every $x, y \in P$.

Example 5.5.5. The Boolean poset B_n is a lattice, with
$$S \wedge T = S \cap T, \qquad S \vee T = S \cup T.$$
The divisor poset $D(m)$ is also a lattice, with
$$a \wedge b = \gcd(a, b), \qquad a \vee b = \operatorname{lcm}(a, b).$$

Remark 5.5.6. A meet-semilattice need not be a lattice. For example, the poset $B_n \backslash \{[n]\}$ is a meet-semilattice but not a join-semilattice when $n \geq 2$.

Definition 5.5.7 (Least and Greatest Elements)**.** An element $\hat{0} \in P$ is the *least element* if $\hat{0} \leq x$ for every $x \in P$. Similarly, $\hat{1} \in P$ is the *greatest element* if $x \leq \hat{1}$ for every $x \in P$.

If P is a finite nonempty lattice, then both $\hat{0}$ and $\hat{1}$ exist and are unique.

Proposition 5.5.8. If a finite meet-semilattice P has a greatest element $\hat{1}$, then P is a lattice.

Proof. Fix $x, y \in P$. Since $\hat{1}$ is an upper bound of both x and y, the set

$$S := \{z \in P : x \leq z,\ y \leq z\}$$

is nonempty. Since P is finite, write $S = \{z_1, \ldots, z_r\}$ and define

$$j := z_1 \wedge z_2 \wedge \cdots \wedge z_r.$$

Since x and y are lower bounds of every element of S, the definition of meet gives $x \leq j$ and $y \leq j$. Hence $j \in S$. Moreover, $j \leq z$ for every $z \in S$. Thus j is the least upper bound of x and y, so

$$x \vee y = j.$$

Therefore every pair of elements has both a meet and a join, and P is a lattice. □

5.6 Subspace Lattices and Order Ideals

Example 5.6.1 (Subspace Lattice)**.** Let $q = p^m$ be a prime power, and let $B_n(q)$ denote the set of all linear subspaces of $\mathbb{F}_q^n$, ordered by inclusion:

$$U \leq W \iff U \subseteq W.$$

This poset is a lattice, with

$$\boxed{U \wedge W = U \cap W, \qquad U \vee W = U + W,}$$

where $U + W := \{u + w : u \in U,\ w \in W\}$ is the sum of the two subspaces.
The poset $B_n(q)$ is graded, with rank function

$$\rho(U) = \dim U.$$

Since the number of k-dimensional subspaces of $\mathbb{F}_q^n$ is the Gaussian binomial coefficient $\binom{n}{k}_q$, as established in Chapter 4,

$$\boxed{F(B_n(q), z) = \sum_{k=0}^{n} \binom{n}{k}_q z^k.}$$

Definition 5.6.2 (Order Ideal)**.** Let P be a poset. A subset $I \subseteq P$ is called an *order ideal* if it is closed downward under the order relation:

$$\boxed{x \in I, \quad y \leq x \quad \Longrightarrow \quad y \in I.}$$

In other words, whenever an order ideal contains an element, it must also contain every element below it.

Example 5.6.3. Consider the chain $P = \{a < b < c\}$. Its order ideals are

$$\varnothing, \qquad \{a\}, \qquad \{a, b\}, \qquad \{a, b, c\}.$$

For example, $\{b, c\}$ is not an order ideal, since $b \in \{b, c\}$ and $a < b$, but $a \notin \{b, c\}$.

Definition 5.6.4 (Poset of Order Ideals)**.** Let $J(P)$ denote the set of all order ideals of P, ordered by inclusion:

$$J(P) := \{I \subseteq P : I \text{ is an order ideal}\}, \qquad I_1 \leq I_2 \iff I_1 \subseteq I_2.$$

Proposition 5.6.5. For any poset P, the poset $J(P)$ is a lattice. For any $I_1, I_2 \in J(P)$,

$$\boxed{I_1 \wedge I_2 = I_1 \cap I_2, \qquad I_1 \vee I_2 = I_1 \cup I_2.}$$

Proof. First, both $I_1 \cap I_2$ and $I_1 \cup I_2$ are order ideals. Indeed, if $x \in I_1 \cap I_2$ and $y \leq x$, then $y \in I_1$ and $y \in I_2$, so $y \in I_1 \cap I_2$. Similarly, if $x \in I_1 \cup I_2$, then x belongs to at least one of I_1, I_2, and every $y \leq x$ belongs to the same ideal. Thus $y \in I_1 \cup I_2$.
Under inclusion, $I_1 \cap I_2$ is the greatest order ideal contained in both I_1 and I_2, while $I_1 \cup I_2$ is the least order ideal containing both. Hence they are the meet and join, respectively. □

Remark 5.6.6. The least and greatest elements of $J(P)$ are $\hat{0} = \varnothing$ and $\hat{1} = P$, respectively.